

Nano Button v3 — React Native Handoff

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Overview

Button v3 is the finalized button: the **dark (primary)** button is from Figma [2nd variant](#) and the **white (secondary)** button is from Figma [Button design](#). Both share one press motion and one haptic.

Variant	Role	Figma source	Rest	Pressed
Dark	Primary action (Continue, Yes, use this email)	2nd variant 1105:19282	1105:19232	1105:19256 (drawn by the designer)
White	Secondary action beside a primary (Back, Edit email)	Button design 1088:18515	1088:18515	dev -> design 1096:19130 (built in the app)

Used on: the onboarding CTA, Skin select, Share, and every parent-flow page (Continue; Back + Continue on Email ID).

Reference implementation (iOS): `GyroQR/GlassButton.swift` (dark faces) and `GyroQR/SlabDome.swift` (white faces, `DomeButtonStyle`, press motion, haptics).

Visual tokens

Both variants are a 16pt-corner rounded rectangle with a 2pt inside stroke. Blur values below are Figma's (a CSS blur); in Skia or any Gaussian API use **$\sigma = \text{blur} / 2$** .

Shared geometry

Token	Value
Height	52pt (Figma, lab); 56pt in the onboarding CTA. Surface stretches to the label's frame
Width	327pt in a 351pt bar (12pt padding); full width of its slot in app
Corner radius	16pt, continuous (squircle) corners
Stroke	2pt, inside, linear gradient top → bottom

Token	Value
Fill inset	Fill stops 1pt in (half the stroke) so its anti-aliased edge cannot show outside the stroke
Gap between stacked pair	12pt
Label	Noontree SemiBold 16, tracking 0 (Figma). Dark: #FFFFFF. White: #1D2539

Dark button (primary) — Figma 2nd variant

Layer	Rest	Pressed
Fill	#212121 solid	Radial #424242 (centre) → #212121 (edge). Centre at 50% x, 14.1% y; radius 57.5% of width across, 130.6% of height down
Stroke (2pt)	#EFEFEF → #D2D2D2; gradient starts 7.9% above the top edge, ends at the bottom edge	#D2D2D2 → #8A8A8A; same gradient line
Inner shadow 1	#D2D2D2 at 50%, x 0, y +4, blur 8, spread 0	#000000 at 50%, x 0, y 0, blur 10, spread 4
Inner shadow 2	#000000 at 100%, x 0, y -9, blur 9, spread 0	—
Drop shadow	none	#000000 at 15%, x 0, y +4, blur 5, spread 0
Label opacity	100%	100%

White button (secondary) — Figma Button design, pressed state built in app

Layer	Rest	Pressed
Fill	#FFFFFF	#F3F4F7
Stroke (2pt)	#FFFFFF → #DBDFE6, top → bottom	unchanged (#FFFFFF → #DBDFE6)
Inner shadow 1	#989FB3 at 10%, y +4, blur 4	#989FB3 at 21%, y +1.5, blur 3 (contact line along the top)

Layer	Rest	Pressed
Inner shadow 2	#989FB3 at 22%, y -6, blur 6	#FFFFFF at 100%, y -3, blur 6 (bounce light along the bottom)
Drop shadow	none	none
Label opacity	100%	80%

The white pressed 21% is #989FB3 at 35% × a 0.6 tuning factor, baked in. Inner shadows draw over the fill and under the stroke.

Disabled (both): no surface at all. The call site draws a flat #F2F3F7 fill, #989FB3 label and a 1pt #989FB3 at 45% border; no press, no haptic.

Motion timesheet

Press-down starts on touch-down (`onPressIn`) and release on touch-up or when the finger leaves (`onPressOut`). Everything starts at $t = 0$ together; nothing is staggered. The action fires on release inside the button, as usual.

Press-down (rest → pressed)

#	Track	Property	From → To	Start	Duration	Curve
1	Haptic	press tap	—	0 ms	instant	see Haptic specs
2	Whole button	scale (centre anchor)	1.000 → 0.980	0 ms	~160 ms to settle	spring S1
3	Whole button	translateY	0 → +0.5pt (down)	0 ms	~160 ms to settle	spring S1
4	Pressed face	opacity (crossfade over the rest face)	0 → 1	0 ms	20 ms	ease-in-out
5	Label (white only)	opacity	1.0 → 0.8	0 ms	20 ms	ease-in-out

Release (pressed → rest)

#	Track	Property	From → To	Start	Duration	Curve
1	Haptic	release tap	—	0 ms	instant	see Haptic specs
2	Whole button	scale	0.980 → 1.000	0 ms	~160 ms to settle	spring S1
3	Whole button	translateY	+0.5pt → 0	0 ms	~160 ms to settle	spring S1
4	Pressed face	opacity	1 → 0	0 ms	20 ms	ease-in-out

#	Track	Property	From → To	Start	Duration	Curve
5	Label (white only)	opacity	0.8 → 1.0	0 ms	20 ms	ease-in-out

A release mid-press retargets from the current value; springs carry their velocity, the 20 ms crossfade just reverses.

Spring S1 (scale + translateY)

Parameter	Value
Design values	response 0.20 s, bounce 0.06
SwiftUI	<code>.spring(response: 0.2, dampingFraction: 0.94)</code> (damping = 1 – bounce)
Physics (mass 1)	stiffness 986.96, damping 59.06
Reanimated	<code>withSpring(v, { mass: 1, stiffness: 986.96, damping: 59.06 })</code>
Overshoot	0.02% (none visible); within 1% of target after ~156 ms

Physics from response T and damping ratio ζ : stiffness = $(2\pi / T)^2$, damping = $4\pi\zeta / T$.

Crossfade curve — ease-in-out, 20 ms (about one frame at 60 fps): `withTiming(v, { duration: 20, easing: Easing.inOut(Easing.ease) })`. Draw both faces stacked and fade the pressed one; never animate the colours themselves.

Haptic specs

Two taps per press: a heavy one at full strength the moment the finger lands, and a light, soft one on release. Both variants use the same haptics.

Event	When	iOS (shipped)	Android equivalent
Press	touch-down, same frame as the visual press	<code>UIImpactFeedbackGenerator(style: .heavy), impactOccurred(intensity: 1.0)</code>	<code>VibrationEffect.createPredefined(EFFECT_HEAVY_CLICK)</code> ; fallback <code>HapticFeedbackConstants.VIRTUAL_KEY</code>
Release	touch-up or finger leaves	<code>UIImpactFeedbackGenerator(style: .light), impactOccurred(intensity: 0.4)</code>	<code>VibrationEffect.createPredefined(EFFECT_TICK)</code> ; fallback <code>HapticFeedbackConstants.KEYBOARD_RELEASE</code> (API 27+)

Rules

- Fire on touch-down, not on the action. The release tap fires even when the press is cancelled by dragging out.
- Disabled buttons fire nothing.
- Respect the app-wide haptics switch: both taps are skipped when it is off.
- Keep the generators warm: iOS calls `prepare()` on both generators at launch and again after each tap, so the first press has no lag.

React Native

- `expo-haptics`: `Haptics.impactAsync(ImpactFeedbackStyle.Heavy)` on press, `ImpactFeedbackStyle.Light` on release. It has no intensity parameter, so the release is slightly stronger than 0.4 on iOS.
- `react-native-haptic-feedback`: `trigger('impactHeavy')` / `trigger('impactLight')`, same limit.
- For an exact match on iOS, a small native module that calls `impactOccurred(intensity:)` with 1.0 and 0.4.

Not shipped: the Button lib also offers a Core Haptics transient (intensity, sharpness 0.75). The app uses the Impact generator above.

React Native implementation notes

1. **Render the faces with `@shopify/react-native-skia`.** RN's `shadow*` / `elevation` cannot draw inner shadows or a 2pt gradient stroke. Draw each face in a `Canvas` sized to the button: `RoundedRect` fill (radial or solid) → inner shadows → stroked `RoundedRect` with a `LinearGradient`.
2. **Inner shadows with spread.** Skia's `Shadow inner` has no spread. Draw it by hand: fill the area outside a copy of the shape inset by `spread` and moved by (x, y), blur with `sigma = blur / 2`, clip to the button shape. The dark pressed shadow (blur 10, spread 4) needs this; the rest-state ones (spread 0) can use `<Shadow inner />`.
3. **Two stacked faces, one opacity.** Keep the rest face always drawn and fade the pressed face over it with a shared value (`pressed` 0 → 1). Do not interpolate colours or shadow values.
4. **Motion on the UI thread.** Use `react-native-reanimated` + `Pressable` (or `Gesture Handler Tap`): in `onPressIn` set the spring and timing targets and fire the haptic; in `onPressOut` set them back. Apply `scale` and `translateY` in one `useAnimatedStyle` on the wrapper.
5. **Continuous corners.** Figma and iOS draw squircle corners. Skia's `RoundedRect` is circular, a difference of under 1pt at 16pt radius; acceptable, or use a squircle path if pixel-matching.

6. **Let shadows overflow.** The pressed drop shadow (y +4, blur 5) draws about 7pt below the button. Size the canvas larger than the button, or turn off clipping on the parent.
7. **Accessibility.** `accessibilityRole="button"` , a label from the text, and `accessibilityState={{ disabled }}` . Suggested, not in the iOS build yet: with Reduce Motion on, keep the crossfade and drop the scale and translate.

Tuning values that are live in the iOS lab but fixed for this handoff: press scale 0.980, bounce 0.06, response 0.20 s, depth 0.5pt, light change 20 ms.